

Anthony Qin

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US Permanent Resident

SUMMARY

Results-driven **Full-Stack Software Engineer** with a proven track record of architecting scalable cloud environments, engineering high-availability APIs, and building responsive, interactive user interfaces. Experienced in managing the end-to-end software development lifecycle in high-velocity environments, taking autonomous ownership of complex technical challenges. Adept at leveraging modern frameworks and integrating advanced data pipelines to deliver robust, user-centric solutions.

EXPERIENCE

Founding Cloud Architect and Backend Engineer

December 2025 – Present

UMA

- Operated as the **sole technical architect** in a high-velocity startup environment, single-handedly designing, provisioning, and scaling the entire **AWS cloud ecosystem** from the ground up to support production workloads.
- Drove **rapid, end-to-end product delivery** by simultaneously wearing multiple hats: engineering the centralized **backend API architecture** while concurrently developing the full suite of **Admin, Operator, and User web portals**.
- Spearheaded complex hardware-software integrations, defining **communication protocols** and building **data ingestion pipelines** to ensure reliable, real-time telemetry between the cloud and **Android-based IoT vending machines**.
- Accelerated time-to-market by taking **total autonomous ownership** of the software development lifecycle — spanning cloud infrastructure, backend orchestration, and frontend architecture—consistently shipping cross-domain features at an accelerated pace.

Cloud Engineer

May 2025 - December 2025

HOOPP (Healthcare of Ontario Pension Plan)

- **Developed foundational cloud automation solutions** across **AWS** and **Azure** to streamline infrastructure management, including provisioning, configuration, and large-scale migrations, automating processes for nearly **400 virtual machines**.
- Engineered **monitoring** and **observability** solutions for **100+ cloud applications**, improving error alert accuracy by over 90% through automated health checks and custom monitoring workflows.
- **Developed an Azure DevOps automation pipeline** that detects CMDB changes and updates legacy XML process templates, reducing a **45-minute manual workflow to under 5 minutes** and eliminating **100%** of manual edits across multiple teams.

Machine Learning Intern

May 2025 - September 2025

Themis AI

- Architected a full-stack, **interactive image segmentation and classification platform** leveraging CAPSA and ActiveLab, optimizing the **active learning data-labeling pipeline** for complex machine learning models.
- Designed and deployed robust **API integrations** to orchestrate the automated fetching of uncertain edge-cases, accelerating **continuous model retraining** and establishing a seamless feedback loop.
- Overhauled the annotation workflow architecture, significantly reducing **data transfer latency** between the frontend and backend while driving higher usability and platform responsiveness.

Undergraduate Researcher

January 2025 - May 2025

University of Toronto / Diamond Schmitt

- Engineered a **constraint-based procedural generation engine** utilizing the **Hierarchical Wave Function Collapse** algorithm to dynamically synthesize complex architectural floor plans.

- Presented research findings and a functional prototype to **Diamond Schmitt Architects**, securing enthusiastic buy-in and establishing a foundation for future technical collaborations.

Software Engineer, HCI & AI

May 2024 - August 2024

Autodesk Research

- Architected and developed **Story Visualizer**, an innovative **LLM-powered tool** utilizing **Next.js**, **LangChain**, and **React Flow** to dynamically extract and visualize narrative elements on an infinite canvas.
- Spearheaded the **end-to-end design and implementation** of the web-based prototype, taking full ownership of technical decisions and working autonomously to translate Research Scientists' visions into a functional product.
- Engineered a highly **flexible and scalable software architecture**, enabling rapid iteration, seamless integration of AI-generated imagery, and agile adaptation to evolving research requirements.

Full Stack Engineer

January 2023 - April 2023

Manulife

- Engineered a comprehensive **full-stack internal dashboard** leveraging **Spring Boot** and **React.js**, streamlining cross-departmental data visibility and drastically reducing manual coordination overhead.
- Developed and deployed robust **PowerShell** and **C automation scripts**, optimizing enterprise workflows and accelerating operational efficiency across technical teams.
- Administered and configured remote server environments via **PuTTY**, ensuring high availability, optimal application performance, and seamless deployment cycles.

PROJECTS

Steam Oracle - AI Game Discovery Engine

- Architected **Steam Oracle**, an **LLM-powered natural language search engine** for video game metadata, delivered as a client-facing prototype for **Blue Ocean Games**.
- Engineered a scalable **Python backend** utilizing **FastAPI** and **LangChain** to orchestrate complex LLM workflows and semantic retrieval pipelines.
- Developed a highly responsive **Next.js frontend** supporting interactive search, dynamic result ranking, and iterative query refinement to validate product-market fit.

Mirror's Memento - Unity Game

- Co-developed a **Unity-based 3D puzzle game** across a 12-person agile team (Doppelganger Studios), successfully showcasing the project at the **CNE Gaming Garage and Innovation Garage**.
- Engineered core gameplay mechanics and systems, including **dynamic controller support**, **grid-based movement algorithms**, and **AI-driven hint systems**.
- Streamlined the asset integration pipeline and resolved complex Git merge conflicts, ensuring continuous delivery for the open beta release on itch.io.

TECHNICAL SKILLS

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Terraform, Jenkins, CI/CD, Linux, PowerShell

Backend & APIs: Python, TypeScript, Node.js (Express), FastAPI, Spring Boot, GraphQL, PostgreSQL, MongoDB, SQL

AI & Frontend: LangChain, React.js, Next.js, Tailwind CSS, NumPy, Pandas, Material UI, React Flow

Game Dev & Core: C#, C, Java, Unity, Godot, Object-Oriented Programming (OOP), Test-Driven Development (TDD), Agile

EDUCATION

Honors Bachelor of Science, Computer Science Specialist

University of Toronto

Toronto, Canada

2021 - 2026